

Assignment-4

1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

❖ GPU Information

Item	Details
GPU Brand	Intel
GPU Model	Intel UHD Graphics 620
GPU Type	Integrated GPU

➤ Short Explanation

The system has an **Intel UHD Graphics 620** GPU. It is an **integrated GPU**, which means it is built into the processor (CPU). It is suitable for web browsing, watching videos, Microsoft Office, and light gaming.

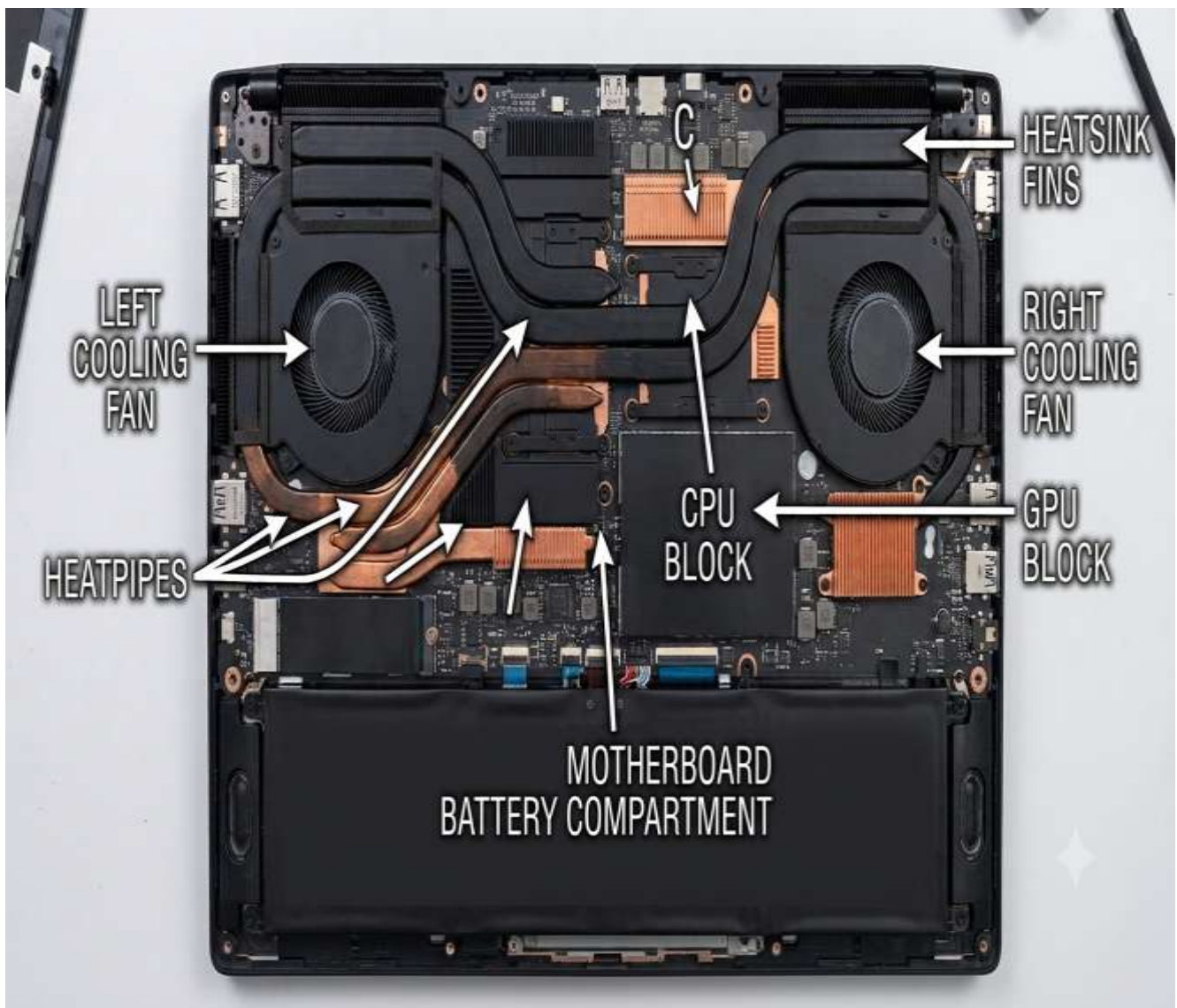
❖ Alternative Example (Dedicated GPU)

Item	Details
GPU Brand	NVIDIA
GPU Model	GeForce GTX 1650
GPU Type	Dedicated GPU

➤ **Explanation:**

The **NVIDIA GeForce GTX 1650** is a **dedicated GPU**. It has its own memory (VRAM) and provides better performance for gaming, video editing, and 3D design.

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



❖ Laptop Cooling Components Functions

- **Cooling Fans (Left & Right Fan):** Spin fast to push hot air out through the laptop vents and pull cool air inside.
- **Heatpipes:** Copper tubes that transfer heat away from the processor chips to the heatsink fins.
- **CPU Block & GPU Block:** Metal plates attached directly on top of the processor chips to absorb heat immediately.
- **Heatsink Fins:** Thin metal plates that spread out the heat so the fan airflow can cool them down quickly.

3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling	Liquid Cooling
1. Cooling Performance	Good. Fine for normal gaming and daily use.	Best. Keeps the GPU much cooler during heavy gaming.
2. Cost (Price)	Cheaper. Comes built-in with most graphics cards.	Expensive. Requires buying extra parts like pumps and radiators.
3. Maintenance	Easy. Just clean the fan dust once in a while.	Harder. Needs leak checks and fluid refills (for custom setups).
4. Noise	Louder. Fans spin fast and make noise under heavy load.	Quieter. Fans run slower and smoother.

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

My Choice: Liquid Cooling for the GPU

❖ Reasons:

✓ **Better Heat Control:**

Streaming IPL matches while playing high-graphics games creates heavy heat over long hours. Liquid cooling keeps the GPU temperature much lower than air cooling.

✓ **Smooth Performance:**

Lower temperatures prevent thermal throttling, ensuring smooth gameplay without frame drops or lag during live streams.

✓ **Quieter Operation:**

Liquid cooling fans run at lower speeds, so fan noise will not be picked up by your microphone while streaming.